

تجربہ سری دوم

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حصہ باطانی

↑ Nonideal op-AMP

typical first order system: $H(s) = \frac{K}{\tau s + 1}$

DC Gain of $-10 \rightarrow K = -10$

$y(t) = K(1 - e^{-\frac{t}{\tau}})$ $t=0 \rightarrow y=0$, $t=T_r \rightarrow y = 0.632 K$ or $\lim_{t \rightarrow \infty} y = K$

$\rightarrow e^{-\frac{T_r}{\tau}} = 0.1 \rightarrow -\frac{T_r}{\tau} = \ln 0.1 \rightarrow T_r \approx 2.19 \tau$

$\Rightarrow \tau = \frac{T_r}{2.19} = \frac{2.19 \mu s}{2.19} = 1 \mu s \rightarrow H(s)_{\text{closed}} = \frac{-10}{1 + 10^{-4}s}$

$\frac{H_{\text{open}}}{1 + H_{\text{open}}} = H_{\text{closed}} \rightarrow \frac{-10}{1 + 10^{-4}s} = \frac{H_{\text{open}}}{1 + H_{\text{open}}}$

$\rightarrow H(s)_{\text{open, loop}} = \frac{-100}{s + 10^4}$

2 model order reduction and Response characteristics

$$1. T(s) = \frac{C(s)}{R(s)} = \frac{G(s)}{1 + 0.1 \times G(s)}, \text{ where } G(s) = \frac{11000 K}{s(s+120)^2}$$

$$\rightarrow T(s) = \frac{\frac{11000 K}{s(s+120)^2}}{1 + \frac{1100 K}{(s+120)^2 s}} = \frac{11000 K}{s(s+120)^2 + 1100 K} \quad \begin{matrix} K=120 \\ \uparrow \\ 1210000 \end{matrix} = \frac{1210000}{s(s+120)^2 + 1210000}$$

$$\rightarrow T(s) = \frac{1210000}{s^3 + 240s^2 + (120)^2 s + 1210000}$$

$p(s)$

Poles: $p(s) = 0 \rightarrow s \approx -128, -112, -10$

با حذف دو قطب بسیار دور از محور سن (تقریباً برابر قطب اول هستند) یک تقریب درجه

یک من زنیج داریم:

2.

General first order $T(s) = \frac{\alpha}{s+\beta}$, $\beta = 10$ (قطب در -10)

$$DC \text{ Gain} = T(s) \Big|_{s=0} = \frac{1210000}{121000} = 10$$

$$\rightarrow T(s) \Big|_{s=0} = 10 \rightarrow \frac{\alpha}{\beta} = 10 \rightarrow \alpha = 100$$

تقریب

$$\rightarrow T(s) = \frac{100}{s+10}$$

تقریب

4. $\tau = k = 1000$,

با $k=110$ ، داریم:

$$T(s) = \frac{22000000}{s^3 + 240s^2 + (120)s + 22000000}$$

$P(s)$

Poles: $P(s)=0 \rightarrow s = -240, -10 \pm j30\sqrt{11}$

قطب سوم، بیش از ۵ برابر بزرگتر از قسمت های حقیقی قطب های مختلف است

در نتیجه سیستم باید سیستم مرتبه دو تقریب می خورد

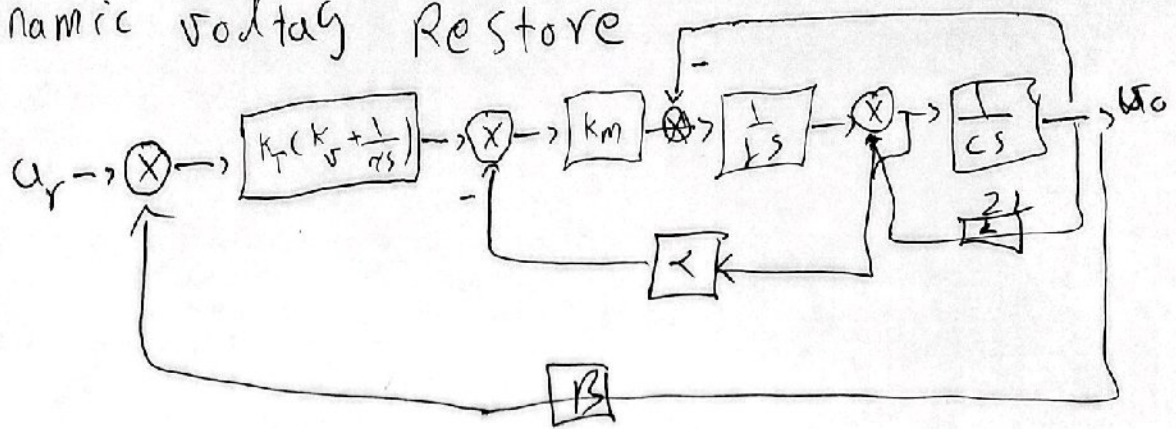
تقریب $T(s) = \frac{\alpha}{s^2 + \beta s + c}$, $s^2 + \beta s + c = 0 \rightarrow s = -10 \pm j30\sqrt{11}$

تقریب $\rightarrow T(s) = \frac{\alpha}{(s - 10 + j30\sqrt{11})(s - 10 - j30\sqrt{11})} = \frac{\alpha}{s^2 + 20s + 10000}$

DC Gain $T(s) = T(s)|_{s=0} = \frac{22 \times 10^6}{22 \times 10^6} = 10 \rightarrow \frac{\alpha}{10000} = 10$

تقریب $\rightarrow \alpha = 10^5 \rightarrow T(s) = \frac{10^5}{s^2 + 20s + 10^4}$

3 Dynamic voltage Restore



1, 2.

$$E(s) = u_r - u_o \beta, \quad H(s) = \frac{u_o}{u_r} \rightarrow u_o = u_r H(s)$$

$$\rightarrow E(s) = u_r - \beta H(s) u_r = (1 - \beta H(s)) / u_r$$

$H(s)$: mason law

$$P_1 = K_T K_m (K_V + \frac{1}{s}) \frac{1}{s} \frac{1}{cs}$$

$$L_1 = -\frac{K_m \alpha}{s}, \quad L_2 = -\frac{1}{cs}, \quad L_3 = -\frac{1}{cc_L s}$$

$$L_4 = -K_T K_m (K_V + \frac{1}{s}) \frac{1}{s} \frac{1}{cs} \beta$$

$$\Delta = 1 - L_1 - L_2 - L_3 - L_4 + L_1 L_2$$

$$\Delta_1 = 1, \quad H(s) = \frac{P_1(s)}{\Delta(s)}$$

$$\rightarrow H(s) = \frac{K_T K_M (K_V + \frac{1}{s}) \frac{1}{CLs}}{1 + \frac{K_M \alpha}{Ls} + \frac{1}{LCLs} + \frac{1}{CCLs} + K_M K_T (K_V + \frac{1}{s}) \frac{\beta}{LCLs}}$$

$$\xrightarrow{xs} H(s) = \frac{(K_T K_M K_V \frac{1}{CL})s + (K_T K_M \frac{1}{\alpha CL})}{s + (\frac{K_M \alpha}{L})s + (\frac{1}{LC} + \frac{1}{CCL} + \frac{K_M K_T K_V \beta}{LC})s + (\frac{K_M K_T \beta}{\alpha LC})}$$

$$\rightarrow \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} s(1 - \beta H(s)) u_r(s)$$

$$\text{step function} \rightarrow u_r(s) = \frac{A}{s} \rightarrow e_{ss} = \lim_{s \rightarrow 0} (1 - \beta H(s)) A$$

$$= 1 - \lim_{s \rightarrow 0} \beta A H(s) = 1 - H(s) \beta A \Big|_{s=0} \quad \begin{matrix} \downarrow \\ \text{constant} \end{matrix} = 1 - 1 = 0$$

$$\text{ramp function} \rightarrow u_r(s) = \frac{A}{s^2}$$

$$\rightarrow \lim_{t \rightarrow \infty} e(t) = 1 - \beta A \frac{H(s)}{s} \Big|_{s=0} > 0$$

$\rightarrow$ type 1 system

4) Hybrid Electric vehicle

$$1. \quad H(s) = \frac{G_{sc}(s) \cdot 0.11(s+0.14)}{s(s+0.1\omega_{VP}) + \omega(s+0.14)(s+0.01901)}$$

$$1 + \frac{G_{sc}(s) \cdot 0.11(s+0.14)}{s(s+0.1\omega_{VP}) + \omega(s+0.14)(s+0.01901)}$$

$$G_{sc} = K_{P_{sc}} \rightarrow H(s) = \frac{K_{P_{sc}} \cdot 0.11(s+0.14)}{s(s+0.1\omega_{VP}) + \omega(s+0.14)(s+0.01901) + K_{P_{sc}}(s+0.14) \cdot 0.11}$$

$$E(s) = R(s) - C(s), \quad \frac{C(s)}{R(s)} = H(s)$$

$$\rightarrow E(s) = (1 - H(s))R(s)$$

$$\text{step input} \rightarrow R(s) = \frac{A}{s} \Rightarrow E(s) = (1 - H(s)) \frac{A}{s}$$

$$\lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} (1 - H(s))A, \quad A=1$$

$$\Rightarrow e_{\text{step}} = 1 - \frac{K_{P_{sc}} \cdot 0.11 \times 0.14}{\omega \times 0.14 \times 0.01901 + K_{P_{sc}} \cdot 0.11 \times 0.14} < 0.01$$

$$\rightarrow \frac{\omega \times 0.14 \times 0.01901}{\omega \times 0.14 \times 0.01901 + K_{P_{sc}} \cdot 0.11 \times 0.14} < 0.01$$

$$\rightarrow 0.0\omega_{VP} < 0.000\omega_{VP} + K_{P_{sc}} + 0.00044$$

$$\rightarrow \boxed{\Delta\omega/\Delta\phi < K_{PSC}} \text{ , it } e_{step} = 0.01$$

$$\text{then } \boxed{K_{PSC} = \Delta\omega/\Delta\phi}$$

$$2. G_{SC}(s) = 100 + \frac{K_{ISC}}{s}$$

$$\rightarrow H(s) = \frac{\left(100 + \frac{K_{ISC}}{s}\right) \left(\frac{0.11(s+0.4)}{s(s+0.0144) + \omega(s+0.4)(s+0.01901)} \right)}{1 + \left(100 + \frac{K_{ISC}}{s}\right) \left(\frac{0.11(s+0.4)}{s(s+0.0144) + \omega(s+0.4)(s+0.01901)} \right)}$$

$$\rightarrow H(s) = \frac{(K_{ISC} + 100s) / (0.11(s+0.4))}{s(s+0.0144) + \omega s(s+0.4)(s+0.01901) + (K_{ISC} + 100s) 0.11(s+0.4)}$$

$$\lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} s E(s) \text{ , } E(s) = (1 - H(s)) R(s)$$

$$R(s) = \frac{A}{s^r} \text{ , } A=1 \rightarrow E(s) = (1 - H(s)) \frac{1}{s^r}$$

$$\rightarrow e_{ramp} = \lim_{s \rightarrow 0} s (1 - H(s)) \frac{1}{s^r} = \lim_{s \rightarrow 0} \frac{1}{s} (1 - H(s))$$

$$= \lim_{s \rightarrow 0} \frac{1}{s} \frac{1}{\left(1 + \frac{K_{ISC}}{s}\right) \left(\frac{0.11(s+0.4)}{s(s+0.0144) + \omega(s+0.4)(s+0.01901)} \right)}$$

$$= \frac{1}{K_{ISC} \frac{0.11 \times 0.4}{\omega \times 0.4 \times 0.01901}} = 4.6\%$$

$$\rightarrow I = \sigma / \omega^2 K I_s + \sigma / \omega^2 \lambda \lambda$$

$$\rightarrow K I_s = \omega \lambda / \lambda \varepsilon \eta$$

5 performance optimization

$$1. H(s) = \frac{\frac{K}{s^2+4s}}{1 + \frac{K}{s^2+4s}} = \frac{K}{s^2+4s+K} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\text{if } \zeta = \frac{\omega}{\sqrt{K}}, \omega_n = \sqrt{K} \quad \text{underdamped} \rightarrow \zeta < 1 \rightarrow \underline{K \geq 9}$$

$$\rightarrow T_s \approx \frac{\xi}{\zeta \omega_n} = \frac{\xi}{\frac{\omega}{\sqrt{K}} \sqrt{K}} = \frac{\xi}{\omega} \approx 1/r/r/r$$

2. for $\underline{K=9}$, T_s is minimum and also $y(t) \leq y_{ct}$

$$\text{if } K=9 \rightarrow H(s) = \frac{K}{(s+\omega)^2} \rightarrow Y(s) = \frac{K}{s(s+\omega)^2}, K=9$$

$$\rightarrow Y(s) = \frac{1}{s} + \frac{-\omega}{(s+\omega)^2} + \frac{-1}{(s+\omega)}$$

$$\xrightarrow{\mathcal{L}^{-1}} y(t) = 1 - e^{-\omega t} - \omega t e^{-\omega t} \rightarrow y(t) \leq 1 = y_{ct}$$

$$3. E(s) = R(s) - Y(s), H(s) = \frac{Y(s)}{R(s)} \quad \text{for } t \geq 0$$

$$\rightarrow E(s) = (1 - H(s)) R(s), H(s) = \frac{K}{s^2+4s+K}, R(s) = \frac{1}{s}$$

$$\rightarrow E(s) = \frac{s+4}{s^2+4s+K}, \int_0^{\infty} e^{ct} dt = \frac{1}{cn} \int_{-\infty}^{\infty} E(j\omega) E(-j\omega) d\omega$$

$$\rightarrow \text{if } K \rightarrow \infty \int_0^{\infty} e^{ct} dt = 0$$